IoT ENABLED NOISE MONITORING SYSTEM

MINI PROJECT REPORT

Submitted By

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to

the APJ Abdul Kalam Technological University in partial fulfillment

of the requirements for the award of the Degree

of

Bachelor of Technology

Electronics and Communication Engineering

Department of Electronics & Communication Engineering
College Of Engineering Muttathara
Thiruvananthapuram-695008
April 2025

DECLARATION

We undersigned hereby declare that the project report “IoT ENABLED NOISE MONITORING SYSTEM", submitted for partial fulfillment of the requirements for the award of the degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a Bona fide work done by us under the supervision of Mr.Saji Justus(Project guide),Mrs.Neena A J and Mrs.Anjali Swapna (project coordinators) . This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the sources. We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma, or similar title of any other University.

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CERTIFICATE

This is to certify that the report entitled 'IoT ENABLED NOISE MONITORING SYSTEM' submitted by **Akhil Richard (CEM22EC004), Amal S Kumar (CEM22EC005), Anumod S Nair (CEM22EC008), Nandhini M (CEM22EC017)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor Of Technology in Electronics & Communication Engineering is a Bona fide record of the project work carried out by him/her under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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This is to certify that the report entitled 'IoT ENABLED NOISE MONITORING SYSTEM' submitted by **Anumod S Nair (CEM22EC008)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor Of Technology in Electronics & Communication Engineering is a Bona fide record of the project work carried out by her under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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This is to certify that the report entitled 'IoT ENABLED NOISE MONITORING SYSTEM' submitted by **Nandhini M (CEM22EC017)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor Of Technology in Electronics & Communication Engineering is a Bona fide record of the project work carried out by him under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ACKNOWLEDGEMENT

We take this opportunity to express our deep sense of gratitude and sincere thanks to all who helped us to complete the work successfully. We are deeply indebted to our guide Mr. Saji Justus Asst. Professor, Department of Electronics and Communication Engineering for his excellent guidance, positive criticism, and valuable comment. It is our privilege to express our sincerest regards to our project coordinator, Mrs. Neena A J for her valuable input, encouragement, and whole- hearted cooperation throughout our project. We deeply express our sincere thanks to our Head of Department Dr RINI JONES.S.B for her support, encouragement, and cooperation. Finally, we thank our parents and friends near and dear ones who directly and indirectly contributed to the successful completion of our project report.

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ABSTRACT

Noise pollution is a significant environmental issue affecting health and productivity in classrooms, industries, and urban areas. Prolonged exposure to high noise levels can cause stress, hearing impairments, and reduced concentration. To address this, an IoT-based noise monitoring system is developed using ESP32 to detect, analyze, and alert users about excessive noise levels in real time. The system integrates a sound sensor to continuously measure noise levels in decibels (dB), with real-time data displayed on an LCD screen. If noise exceeds a predefined threshold (e.g., 75 dB in industries), a buzzer rings, and an alert is sent via the Blynk app for remote monitoring. In classroom environments, when the noise crosses the set limit, the ESP32-CAM captures an image, helping to identify the source of disturbance. Additionally, the system logs noise data and represents it through visual charts, allowing trend analysis for better noise management. With Wi-Fi connectivity, users can remotely monitor noise levels, making it easier to implement corrective measures. This IoT-based noise monitoring system offers an efficient and automated solution for tracking, analyzing, and controlling noise pollution, improving classroom discipline, workplace safety, and overall noise management.

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CHAPTER 1
INTRODUCTION

Noise pollution has become a growing environmental concern in modern urban life, with sources such as traffic, industrial activity, and crowded public spaces contributing to increased noise levels. Prolonged exposure to high noise levels can lead to serious health issues including stress, sleep disturbances, hearing loss, and even cardiovascular problems. In sensitive environments like classrooms, hospitals, and libraries, maintaining a quiet atmosphere is crucial for comfort, concentration, and well-being.

This project aims to address the issue of noise pollution .The system continuously monitors sound levels and alert them.By implementing this system, the project helps promote noise awareness and encourages proactive control of sound disturbances in noise-sensitive areas.

CHAPTER 2
LITERATURE SURVEY

The literature survey on autonomous vehicles utilizing image processing and Raspberry Pi focuses on the integration of computer vision algorithms and hardware platforms for real-time object detection, recognition, and decision-making. Researchers have explored various image processing techniques, such as convolutional neural networks and feature extraction methods, to analyze visual data captured by onboard cameras. The Raspberry Pi, with its low-cost and compact design, has been widely employed as a computational unit for image processing tasks. This survey investigates the advancements in autonomous driving systems that leverage image processing and Raspberry Pi, highlighting their benefits, limitations, and potential areas for improvement.

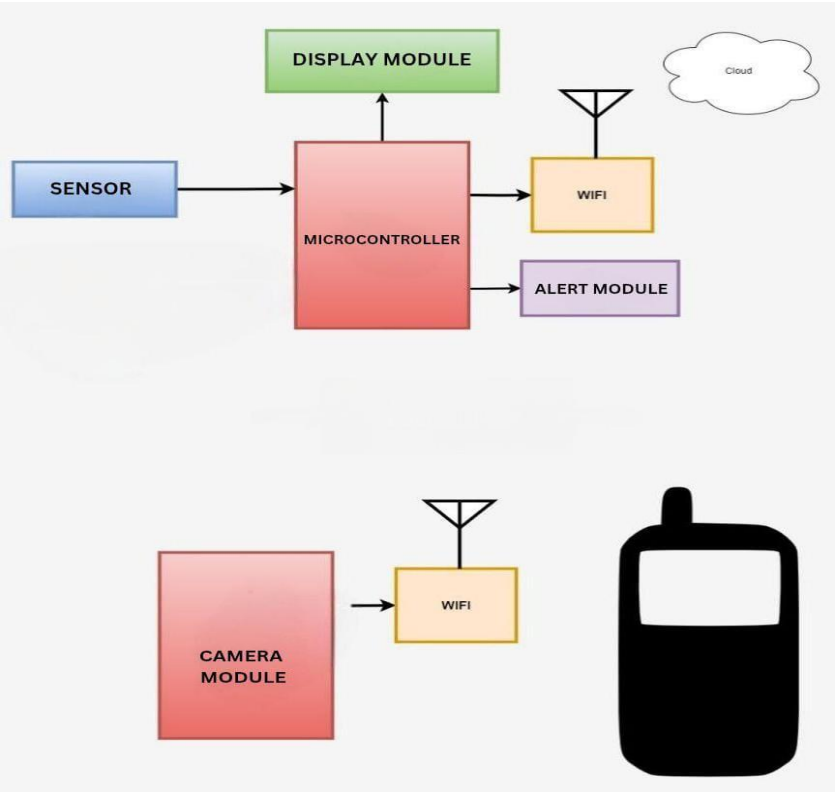
Sl. no	**Title**	**Authors**	**Positive aspects**	**Limitations**
1	IoT Based Real-Time Noise Mapping System for Urban Sound Pollution Study	Sakib Ahmed, Touseef Saleh Bin Ahmed, Sumaiya Jafreen, Jannatul Tajrin, Jia Uddin	Facilitates real-time noise mapping across multiple locations. Employs IoT and web technologies for data visualization.	The system's scalability and data accuracy in high-density urban areas require further exploration.
2	IoT-Enabled Acoustic Sensing Platform for Noise Pollution Monitoring	Gunatilaka Dolvara	Affordable and scalable system design. Reliable performance with low latency and energy consumption.	Limited deployment in dense urban environments with variable noise sources

3	Design of Low-Cost Wireless Noise Monitoring Sensor Unit Based on IoT Concept	Anachkova Maja	Cost-effective and suitable for smart city implementations. Emphasizes granularity in data collection.	Challenges in noise event classification and high-density deployment.
4	IoT Cloud-Based Noise Intensity Monitoring System	Fatema Tashfat	Portable and user-friendly. Effective for urban areas with limited monitoring infrastructure.	Focused primarily on specific urban scenarios without addressing scalability.
5	Air and noise pollution monitoring system	Amulya Gulati Amit Prakash Jayashree. A K. P. Suresh	a wireless sensor network that automatically transmits, processes, analyses, and visualises data The IOT concept is useful in achieving real-time monitoring of sensor data.	Poor sensory data quality requires preprocessing. Lack of robust classification methods for noisy and big data.

6	Smart Noise Pollution Monitoring System	R. Subraja, T. Meyyapan, A. Padmapriya, S. Santhosh Kumar	Real-time noise monitoring with precise data collection. Scalable for larger areas and adaptable to various settings.	Limited discussion on sensor calibration for diverse environments. Lack of detailed evaluation in highly noisy urban settings.

CHAPTER 3
SYSTEM OVERVIEW

3.1 General Block Diagram



- Microcontroller (e.g., Arduino/ESP32): Acts as the brain of the system, handling sensor inputs, processing data, and controlling outputs.

- Sound Sensor (e.g., microphone module like MAX4466 or KY-037): Captures ambient sound and converts it into analog signals.

- Display Unit (e.g., OLED/LCD): Shows the measured sound levels in decibels (dB).

- Wireless Module (optional - WiFi/Bluetooth): For remote monitoring or sending alerts.

- Camera Module (e.g., ESP32-CAM): Captures real-time images or video. It includes an onboard microcontroller (ESP32) with Wi-Fi capability, allowing for both image processing and wireless data transmission.

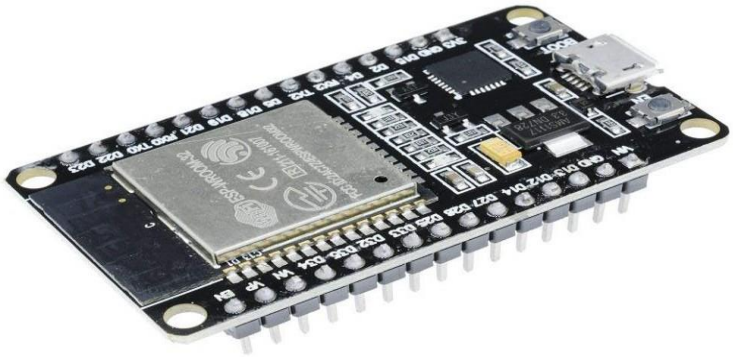
- Alert Module (e.g., Buzzer): Provides audible alerts or notifications when certain conditions are met (e.g., when noise is detected). It is controlled by the microcontroller and activated through digital signals.

CHAPTER 4
HARDWARE COMPONENTS

The main hardware components of our project are

- ESP32
- ESP32 CAM
- Sound Sensor
- LCD Display
- Buzzer
- Power Supply Board
- Connecting wires

4.1 ESP32

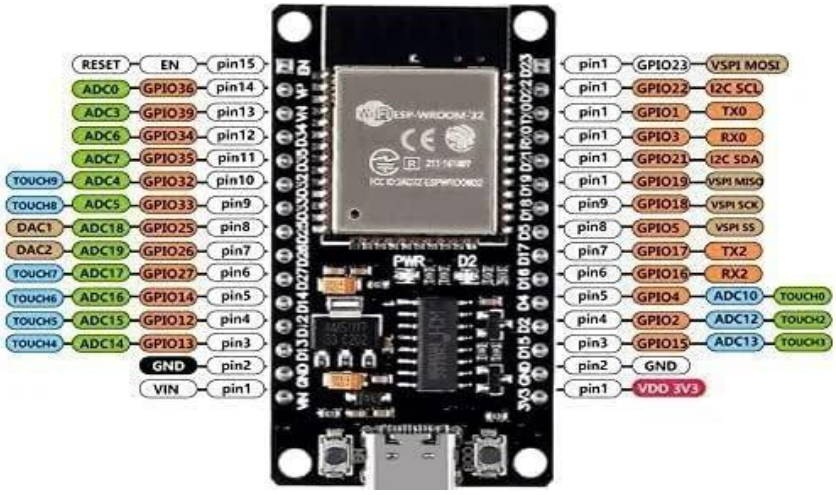


The ESP32 is a powerful and cost-effective microcontroller with built-in Wi-Fi and Bluetooth, making it ideal for IoT, automation, and embedded applications. It offers high performance with a dual-core processor, multiple GPIOs, and support for ADC, DAC, SPI, I2C, UART, and PWM interfaces. Despite its affordability, the ESP32 is highly versatile, used in smart home automation, industrial monitoring, robotics, and AI-powered vision tasks like face recognition (e.g., ESP32-CAM). It supports various programming environments, including Arduino IDE, ESP-IDF, MicroPython, and PlatformIO, catering to both beginners and advanced developers. Additionally, its low power consumption and deep sleep modes make it suitable for battery-powered and energy-efficient applications.

Specifications of the ESP32 :

- CPU: Dual-core Tensilica Xtensa LX6, up to 240 MHz
- RAM: 520 KB SRAM + optional 8 MB PSRAM in some models
- Flash Storage: Typically 4MB (varies by model)
- Wi-Fi: 802.11 b/g/n (2.4 GHz) with AP, Station, and Mesh support
- Bluetooth: Bluetooth 4.2 (Classic + BLE)
- Operating Voltage: 3.3V
- Input Voltage: 5V (via USB) or Vin pin (up to 12V)
- Power Consumption: Active mode: ~160 Ma
  Deep Sleep mode: ~10 µA
- GPIOs: 30 pins total, with up to 25 configurable GPIOs
- Analog Inputs: 16 channels (12-bit ADC)
- Analog Outputs: 2 DAC (8-bit)
- USB-to-Serial Converter: CP2102 or CH340 (depends on the board)
- Reset & Boot Buttons: Available for easy flashing
- Built-in LED: Typically on GPIO2
- Communication Protocols:3 UART ports,2 I2C ports,3 SPI interfaces,2 I2S interfaces (for audio processing)
- Programming & Development Compatible with:Arduino IDE,ESP-IDF (Official SDK),MicroPython & CircuitPython,PlatformIO.

ESP32 PIN DIAGRAM



4.2 ESP32 CAM

The ESP32-CAM is a compact and low-cost microcontroller module with an integrated OV2640 camera, built-in Wi-Fi and Bluetooth, and support for AI-powered applications like face recognition and object detection. It has a small form factor of 27mm × 40mm, making it ideal for space-constrained projects. The camera supports a resolution of up to 1600×1200 pixels, providing decent image quality for IoT and surveillance applications. With high processing power and low power consumption, it is widely used in smart security systems, remote monitoring, home automation, and AI-driven projects. Its ability to store images on a microSD card and stream video over Wi-Fi makes it a powerful yet budget-friendly option for developers.

Features of ESP32-CAM

1. High-Performance Microcontroller & Memory; Powered by the ESP32-S chip with a dual-core Tensilica Xtensa LX6 processor (up to 240 MHz) and 520 KB SRAM; it includes 4MB flash memory and supports 4MB external PSRAM, making it suitable for handling large image data and AI applications.

2. Camera & Image Processing Capabilities; Equipped with an OV2640 camera module supporting resolutions up to 1600×1200 pixels, allowing real-time video streaming and image capture; it supports AI features like face detection and recognition, enabling smart vision-based applications.

3. Wireless Connectivity & Storage Expansion; Built-in Wi-Fi (802.11 b/g/n 2.4 GHz) and Bluetooth 4.2 (Classic & BLE) for remote communication and control; features a microSD card slot supporting cards up to 4GB, enabling local image and video storage.

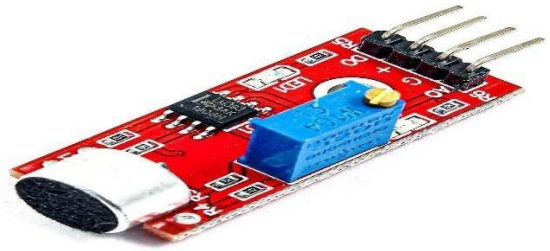
4. Power Efficiency & GPIO Availability; Operates on 3.3V – 5V, with low-power modes for energy-saving applications; provides 9 usable GPIO pins, allowing integration with external sensors, actuators, and other peripherals.

5. Development Support & Remote Updates; Compatible with Arduino IDE, ESP-IDF, MicroPython, and PlatformIO, making programming flexible; supports Over-the-Air (OTA) updates, enabling remote firmware upgrades without the need for physical connections.

6. Applications in IoT & AI-based Projects; Widely used in security and surveillance systems, smart home automation, robotics, and IoT applications; its AI capabilities make it ideal for face recognition, motion detection, and automated monitoring systems.

The ESP32-CAM is a powerful yet affordable camera module with wireless connectivity, AI capabilities, and versatile development support, making it ideal for vision-based IoT and embedded applications.

4.3 **Sound Sensor**



The KY-037 sound sensor is a high-sensitivity module designed to detect sound intensity and noise levels in various environments. It features both analog (A0) and digital (D0) outputs, allowing for real-time audio analysis and threshold-based sound detection.

The sensor includes a built-in microphone and an adjustable potentiometer to set sensitivity levels, making it adaptable to different applications. It operates on a 3.3V–5V power supply, making it compatible with microcontrollers like ESP32 and Arduino.

With its ability to detect and respond to sound changes, the KY-037 is commonly used in noise detection systems, voice-activated controls, security alarms, and home automation projects, such as clap-activated switches.

Applications of Sound Sensor

1. Noise Detection & Security Systems; The KY-037 can monitor noise levels in classrooms, offices, and public spaces, triggering alerts when excessive sound is detected; it is also used in security systems to detect sudden loud noises like glass breaking or unauthorized entry, activating alarms or notifications.

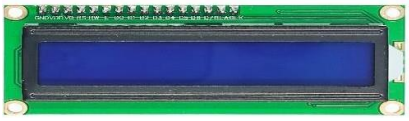
2. Voice-Activated Control & Smart Home Automation; The sensor is commonly used in clap-controlled devices where a sound triggers appliances like lights or fans; it can also be integrated into smart home automation systems to control doors, lights, or security cameras based on specific sound patterns.

3. Industrial Machine Monitoring & Environmental Noise Analysis; In factories and industries, the KY-037 helps detect abnormal machine sounds, preventing failures by signaling maintenance needs; it is also used in urban noise monitoring, measuring pollution levels in high-traffic or construction areas to ensure regulatory compliance.

4. Robotics & AI Interaction; The sensor enhances robotic systems by enabling them to react to sound cues or respond to voice commands, improving interactivity in AI-based systems; it is also integrated into toys, speech-controlled applications, and educational robotics, making automation more interactive and intuitive.

The KY-037 sound sensor is widely used in IoT, security, automation, industry, and AI-based applications, offering an efficient way to detect and respond to sound variations in different environments.

4.4 LCD Display



The 16x2 LCD display is a widely used alphanumeric display module that can show 16 characters per line across 2 rows. It is based on the HD44780 controller, making it compatible with microcontrollers like ESP32, Arduino, and Raspberry Pi. This low-power, cost-effective display supports custom characters, scrolling text, and basic symbols, making it ideal for IoT, automation, and embedded projects. It can be interfaced using either 4-bit or 8-bit parallel communication and often includes an adjustable backlight and contrast control for better visibility in different lighting conditions.

4.5 Buzzer

An active low buzzer is an electronic sound-emitting device that operates when the input signal is pulled low (0V or ground). It contains an internal oscillator, allowing it to generate a continuous sound when activated without requiring an external frequency signal. These buzzers typically operate on 3.3V to 12V, making them compatible with microcontrollers like ESP32, Arduino, and Raspberry Pi. They are commonly used in alarm systems, notifications, timers, and smart automation projects. The active low functionality ensures that the buzzer remains silent at a high signal (logic 1) and produces sound when triggered by a low signal (logic 0), making it useful in circuits where low-level activation is preferred.

4.6 Connecting wires

Jumper wires are electrical wires with connectors at both ends, used to create temporary or permanent connections between components on a breadboard, microcontroller, or circuit board. They come in three types: male-to-male, male-to-female, and female-to-female, allowing flexible connections between different electronic modules. Typically insulated with plastic or silicone, they are available in various lengths and colors for easy circuit identification. Jumper wires are essential in prototyping, debugging, and testing electronic circuits as they eliminate the need for soldering, making them ideal for use with Arduino, ESP32, Raspberry Pi, and other development boards.

CHAPTER 5
SOFTWARE DESCRIPTION

5.1 ARDUINO IDE

The Arduino IDE (Integrated Development Environment) is an open-source software used for writing, compiling, and uploading code to Arduino boards and other compatible microcontrollers like the ESP32 and ESP8266. It provides a simple and user- friendly interface, supporting the C and C++ programming languages with built-in functions for hardware interaction. The IDE includes a code editor, compiler, and serial monitor, allowing developers to write, debug, and communicate with connected devices. It supports libraries and board managers, enabling easy integration of sensors, displays, and modules.
Compatible with Windows, macOS, and Linux, Arduino IDE is widely used for IoT projects, robotics, automation, and embedded systems development.

Key Features of Arduino IDE

1. User-Friendly Interface & Cross-Platform Support; Features a simple, intuitive interface for easy coding and debugging; compatible with Windows, macOS, and Linux, making it accessible to developers across different operating systems.

2. Support for Multiple Boards & Microcontrollers; Natively supports Arduino boards, along with ESP32, ESP8266, and other third-party microcontrollers through the Board Manager, enabling flexible hardware compatibility.

3. Built-in Code Editor & Syntax Highlighting; Includes a text editor with syntax highlighting, auto-indentation, and code suggestions, helping developers write and organize code efficiently.

4. Integrated Compiler & Debugging Tools; Provides a built-in compiler for verifying and compiling code, along with a serial monitor and plotter for real-time debugging and data visualization.

5. Extensive Library Support & Community Contributions; Offers a vast collection of pre-built libraries for sensors, displays, communication modules, and peripherals, with strong community support for additional third-party libraries.

6. Easy Code Uploading & OTA Updates; Allows seamless code uploading via USB, UART, or wireless (OTA) methods, making firmware updates convenient, especially for IoT applications.

7. Open-Source & Expandable; Fully open-source, allowing users to modify and extend its functionality; supports integration with PlatformIO, Visual Studio Code, and other advanced development tools for enhanced programming features.

5.2 BLYNK APP

The Blynk app is a powerful and user-friendly platform designed for building IoT (Internet of Things) applications that allow remote control and monitoring of smart devices. It supports Arduino, ESP32, Raspberry Pi, and other microcontrollers, enabling seamless communication over Wi-Fi, Bluetooth, and cellular networks. The app provides a drag-and-drop interface for creating custom dashboards with widgets like buttons, sliders, graphs, and gauges, making it easy to control hardware without complex coding. Blynk also offers real-time data monitoring, notifications, and cloud storage, making it ideal for home automation, industrial monitoring, and smart IoT solutions. With support for Blynk Cloud and private servers, it ensures flexibility and scalability for both hobbyists and professional developers.

Key Features of Blynk App

1. Cross-Platform Compatibility & Cloud Integration; Works on Android, iOS, and web browsers, allowing users to control IoT devices from anywhere; provides Blynk Cloud for seamless data storage and device management.

2. Drag-and-Drop Dashboard & Custom Widgets; Offers an easy-to-use interface with widgets like buttons, sliders, graphs, and gauges, enabling users to design interactive IoT dashboards without advanced coding.

3. Multiple Connectivity Options & Device Support; Supports Wi-Fi, Bluetooth, Ethernet, and cellular networks; compatible with ESP32, Arduino, Raspberry Pi, and other microcontrollers, making it versatile for IoT applications.

4. Real-Time Data Monitoring & Notifications; Displays sensor readings and device status in real-time; includes push notifications, email alerts, and SMS to notify users of critical events or system updates.

5. Blynk Cloud & Private Server Support; Provides free access to Blynk Cloud for managing devices remotely; allows users to set up private servers for enhanced security and data control.

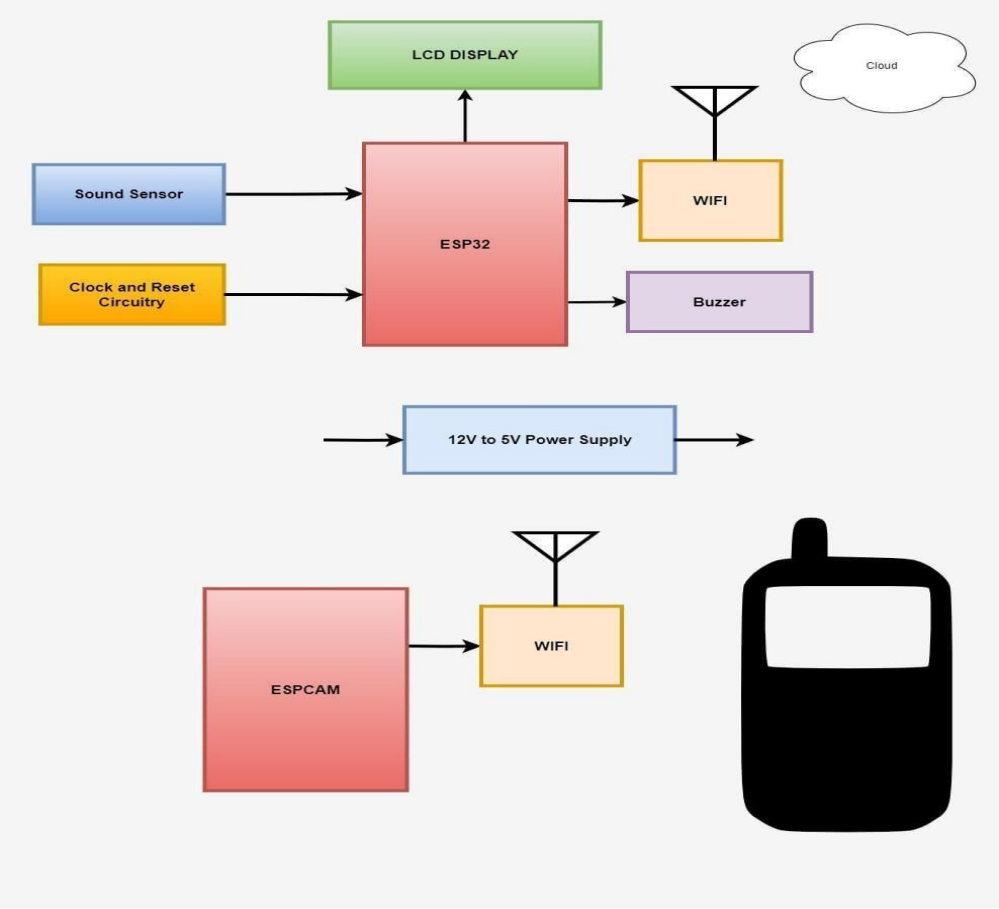
6. OTA (Over-the-Air) Updates & Remote Firmware Management; Enables wireless firmware updates for connected devices, eliminating the need for physical access during software upgrades.

7. Multi-User & Role-Based Access Control; Supports multiple users with role-based permissions, making it suitable for smart home automation, industrial monitoring, and team-based IoT projects.

CHAPTER 6
DESIGN

The design of the IoT-enabled noise monitoring system consists of both hardware and software components working together to detect, analyze, and manage noise levels in real time. The ESP32 microcontroller serves as the core unit, processing data from a sound sensor that continuously measures noise levels in decibels (dB). A LCD display shows real-time noise readings, while a buzzer is triggered when noise exceeds a predefined threshold in industries. For classroom monitoring, an ESP32-CAM captures images when noise levels surpass the limit, helping identify sources of disturbance. The system uses Wi-Fi communication to send alerts via the Blynk IoT platform, enabling remote monitoring.
Additionally, noise data is logged and visualized through graphs for trend analysis. The entire system is programmed using Arduino IDE and designed to enhance classroom discipline and industrial safety through automated noise control and monitoring.

6.1 Block Diagram



6.1.1 Explanation of Block Diagram

1. ESP32 Microcontroller (Central Processing Unit)

The ESP32 is the core component that processes inputs and controls outputs.
It receives data from the sound sensor and decides whether to trigger alerts.
It manages the LCD display, buzzer, and WiFi communication.

2. Sound Sensor

Detects ambient noise levels and sends signals to ESP32 when the noise exceeds a predefined threshold.

3. LCD Display

Displays real-time noise status messages such as "Noise level" and "Noise intensity".

4. Buzzer

Activates when noise exceeds the threshold to provide an audible alert.

5. WiFi Module (ESP32 and ESP32-CAM)

The ESP32 transmits data via WiFi to the cloud or a mobile interface for remote monitoring. The ESP32-CAM also has a WiFi module for streaming live video or images.

6. ESP32-CAM

Captures images or video footage when noise is detected. Sends
the recorded data via WiFi for remote viewing.

7. Power Supply (12V to 5V Converter)

Converts 12V power to 5V for ESP32 and ESP32-CAM operation.

8. Cloud Storage / Mobile Interface

Receives noise alerts and visual data via WiFi.
Allows users to monitor noise levels and images remotely via a smartphone or web interface.

6.2 Working

1. Power Supply & Initialization

- The system is powered by a 12V to 5V converter, ensuring that ESP32 and ESP32-CAM operate at the required voltage.
- Once powered on, the ESP32 microcontroller initializes all connected components (sound sensor, LCD display, buzzer).

2. Noise Detection

- The sound sensor continuously monitors the surrounding noise levels.
- When the noise level exceeds a predefined threshold (set in the ESP32 program), the sensor sends a signal to the ESP32.

3. Processing the Noise Data

- The ESP32 receives the signal from the sound sensor and determines whether the noise is excessive.
- If the noise is within an acceptable range, the system continues monitoring without triggering an alert.
- If the noise exceeds the threshold, the system triggers alert mechanisms.

4. Alert Mechanisms

- The LCD display shows a warning message, such as "Noise level" and "Noise intensity".
- The buzzer is activated to provide an audible alert to notify users of excessive noise.
- The ESP32-CAM captures an image or records a short video of the environment when noise is detected.

5. WiFi Data Transmission

- The ESP32 transmits the noise data and alerts to a remote cloud server or mobile interface via WiFi.
- The ESP32-CAM also transmits captured images/video via WiFi, allowing users to remotely monitor the noisy environment.

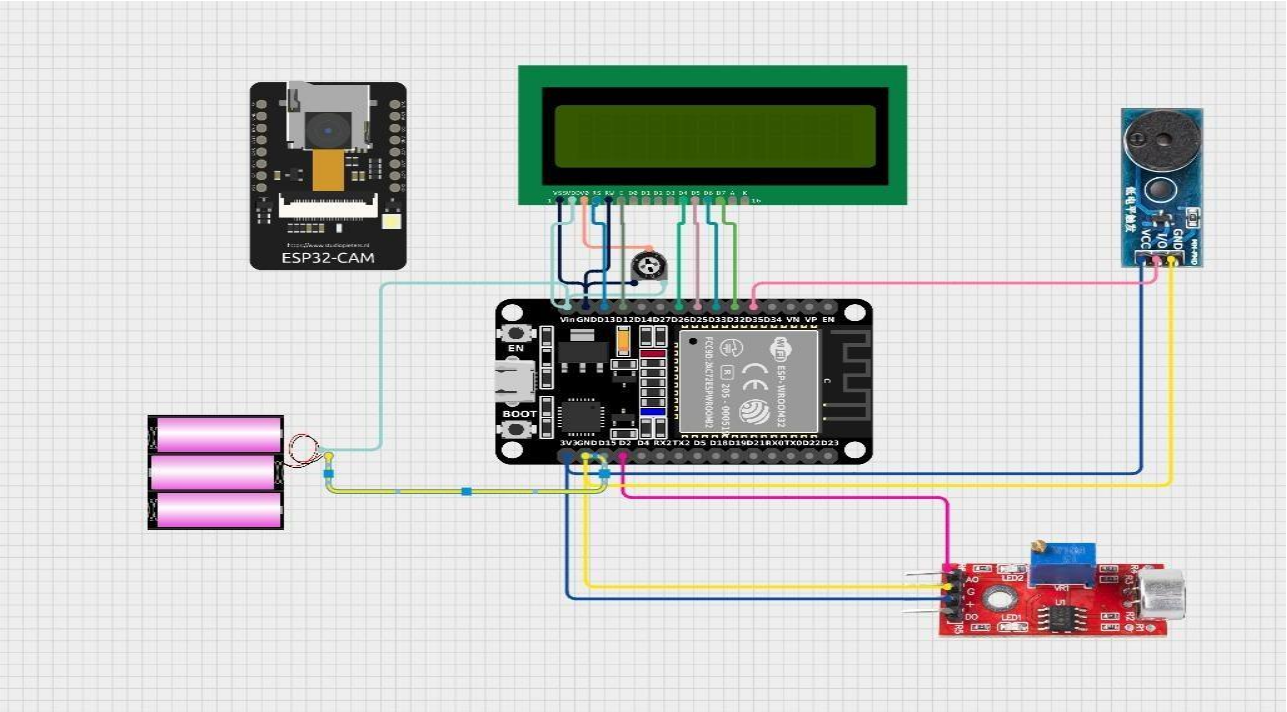
6. Remote Monitoring & Data Storage

- Users can view real-time noise alerts and captured images on a mobile device or cloud dashboard.
- This allows authorities (e.g., teachers in a classroom) to take necessary action if noise pollution persists.

7. Continuous Monitoring

- The system continues monitoring noise levels.
- The ESP32 ensures the system remains in continuous operation, detecting and reporting noise levels in real-time.

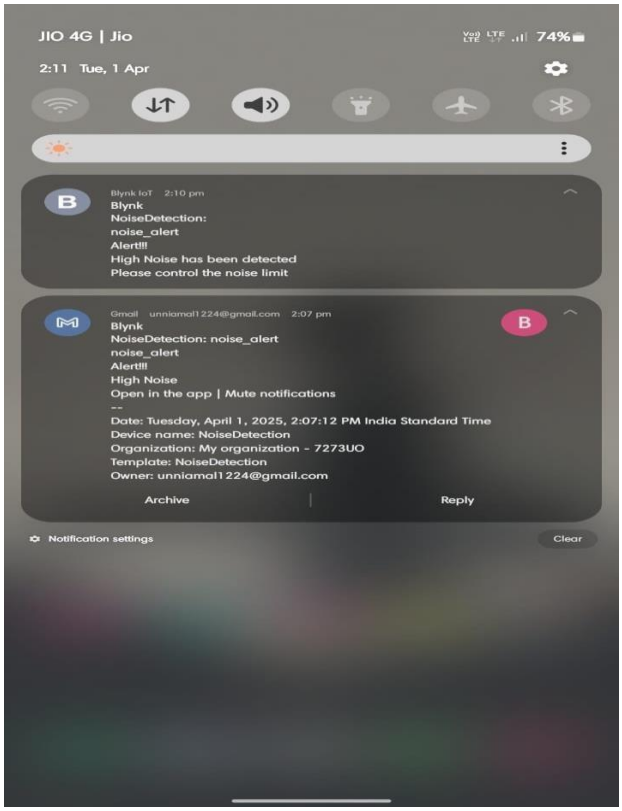
CHAPTER 7
CIRCUIT DIAGRAM

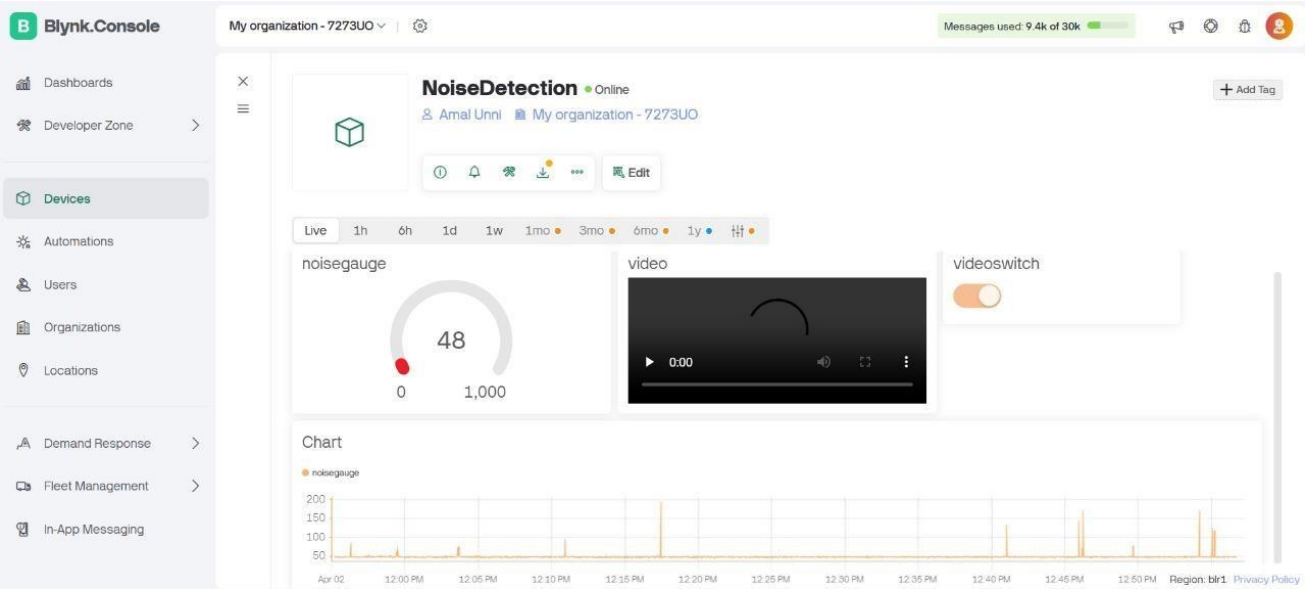


This circuit diagram represents a noise monitoring system using an ESP32 microcontroller. The ESP32 acts as the central unit, interfacing with sensors and modules. An ESP32-CAM module is included to capture images or stream video for surveillance purposes. A 16x2 LCD display is connected to the ESP32 to show real-time sensor data or system status, with a potentiometer used to adjust its contrast. A sound sensor is used to detect loud noises or disturbances in the environment. A power is supplied. This setup is ideal for basic noise monitoring applications, providing both audio and visual monitoring capabilities along with a user-friendly display interface.

CHAPTER 8
RESULT AND ANALYSIS

8.1 **Results**





The IoT Enabled Noise Monitoring System was successfully designed and implemented using ESP32 and ESP32-CAM. The system effectively detects excessive noise levels using a sound sensor, displays alerts on an LCD, and triggers a buzzer when the noise crosses a set threshold. Additionally, the ESP32-CAM captures images or video during noisy events and transmits them via WiFi for remote monitoring. Despite minor challenges related to sensor calibration, power stability, and WiFi connectivity, the system performed reliably in real-time scenarios. It proves to be a useful tool for monitoring and controlling noise pollution in environments such as industries,classrooms,libraries,offices etc.

CHAPTER 9
CHALLENGES FACED AND FUTURE SCOPE

9.1 Challenges Faced

- Camera Performance: ESP32-CAM struggled in low-light conditions, affecting image quality.

- Component Integration: Synchronizing the operation of the LCD, buzzer, sensor, and camera with the ESP32 was tricky.

- WiFi Connectivity: Frequent disconnects affected real-time data transmission and image upload.

9.2 Future Scope

- Integrate with municipal systems to monitor noise pollution in urban areas and assist in enforcing regulations.

- If unusual or loud noises (like shouting, glass breaking) are detected, trigger an emergency alert for security purposes.

- Connect with smart traffic systems to monitor and control honking or loud vehicles in restricted zones.

Chapter 10
CONCLUSION

The development of the Noise Pollution Detection System successfully demonstrates how IoT technology can be applied to monitor and control noise levels in sensitive environments. By integrating ESP32, ESP32-CAM, sound sensors, and wireless communication, the system provides real-time alerts and visual monitoring, ensuring prompt response to excessive noise. This project not only helps in maintaining a quiet atmosphere in places like industries,offices, classrooms and libraries but also promotes awareness about noise pollution. With further enhancements, such systems can be expanded for use in public areas and smart city applications.

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